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*Geology* 1982;10;50-54
doi: 10.1130/0091-7613(1982)10<50:PRFNME>2.0.CO;2

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# Paleomagnetic results from northeastern Mexico: Evidence for large Mesozoic rotations

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## ABSTRACT

**Preliminary paleomagnetic evidence indicates that a significant part of Mexico rotated 130° counterclockwise during early Mesozoic time, coeval with the opening of the Gulf of Mexico, and that a major structural discontinuity must exist between Mexico and cratonic North America. Measurements made on samples from fourteen isolated outcrops of the Huizachal Group, a red-bed sequence exposed in northeastern Mexico indicate that these sediments were deposited in rifted basins and could range in age from post–Middle Permian to pre–Early Cretaceous. Of samples analyzed from the overlying Zuloaga Limestone (Oxfordian) and the Cuesta del Cura Formation (Aptian-Albian), the oldest yield a paleomagnetic pole position that implies that the sampling area was originally oriented about 130° clockwise relative to stable North America. Progressive counterclockwise rotation continued to Middle Jurassic time. Because of the limited geographical distribution of sampling sites, it is not possible to assess whether most of Mexico shared this rotation or whether the proposed left-lateral motion along major west-northwest–trending faults caused the rotation of a smaller crustal block. In either case, our results strongly suggest a major structural discontinuity in northern Mexico.**

## INTRODUCTION

The Huizachal Group is a lower Mesozoic red-bed sequence in northeastern Mexico that was deposited during the formative stages of the Gulf of Mexico. Initially, paleomagnetic samples were collected at several isolated outcrops to obtain stratigraphic information. This was partially achieved, but a more significant result was the discovery that the Mexican pole positions differ greatly from the North American pole positions (Belcher, 1979). In a second field season we expanded the initial sample collection. We present here the results from the combined data set and draw attention to their possible tectonic significance.

### Geologic Setting

Strata identified as the Huizachal Group are the oldest Mesozoic sediments exposed in the Gulf of Mexico region. These red beds occupy a relatively large expanse of geologic time that could range from post–Middle Permian to pre–Early Cretaceous. The Huizachal red beds uncomformably overlie upper Paleozoic metamorphic and sedimentary rocks and are both conformably and unconformably overlain by Upper Jurassic to Lower Cretaceous limestones and evaporites.

Outcrops of the Huizachal red beds are scattered across a 300,000 km² area of northeastern Mexico. In most instances, exposures are small and of poor quality, and detailed stratigraphic and structural relationships between exposures are not known. In the Ciudad Victoria area, in the state of Tamaulipas, the presence of an unconformity within the red beds has led geologists to divide the Huizachal Group into the La Boca and overlying La Joya Formations (Mixon, 1963). Red beds in the Torreon area have been assigned to the Nazas Formation (Pantoja-Alor, 1963; Cordoba, 1964; Clemons, 1965; Clemons and McLeroy, 1965). The Huizachal red beds are devoid of fossils except for one exposure of the La Boca Formation near

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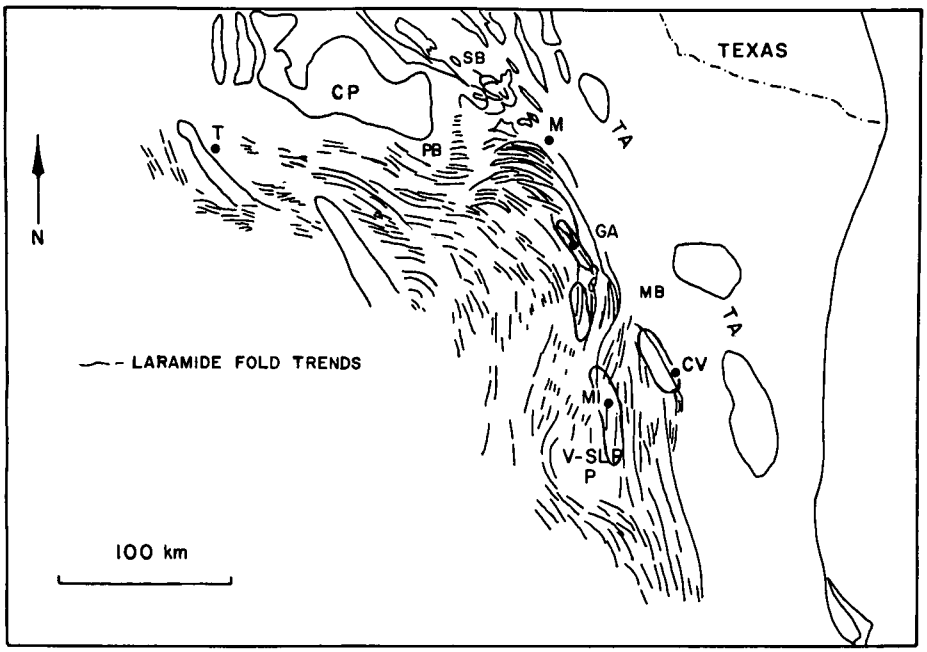



**Figure 1. Structural elements and Laramide fold trends in northeastern Mexico. CP = Coahuila Peninsula; TA = Tamaulipas Arch; V-SLP = Valles–San Luis Potosi Platform; PB = Parras Basin, SB = Sabinas Basin; MB = Magiscatzin Basin; M = Monterrey, T = Torreon, GA = Galeana; CV = Ciudad Victoria, MI = Miquihuana. Sampling site 1 is near Miquihuana, site 12 is near Galeana, and all other sites are in the Ciudad Victoria area.**

Ciudad Victoria which contains Late Triassic plant fragments (Mixon and others, 1959).

Red beds of the Huizachal Group were apparently deposited as graben fill by vegetated low-gradient alluvial fans within an irregular block-faulted terrain (Belcher, 1979). The exposed thickness of the red beds varies from about 2,000 m to less than 10 m. The red beds are characterized by alternating conglomeratic and nonconglomeratic sedimentary packages that vary in thickness from 50 to 400 m.

The Huizachal Group is overlain—usually unconformably—by marine rocks of various ages. This suggests a gradual subsidence of an irregular topography during the Mesozoic. In the Huizachal and Peregrina Canyons near Ciudad Victoria, the red beds are overlain with angular unconformity by the Oxfordian Zuloaga Limestone; samples were taken at both localities. In the Peregrina Canyon, we also collected samples of the Aptian-Albian Cuesta del Cura Formation. The sampling areas are shown in Figure 1.

### Experimental Results

The oriented core samples were stored in a magnetically shielded room (< 100 nT) for at least several weeks prior to measurement and remained inside the room throughout the experimental process. The clastic rocks were thermally demagnetized at 200, 300, 400, 500, and 600 °C and in most cases also at 625 and 650 °C. The carbonate rocks were demagnetized in 50° increments from 200 to 450 °C. The magnetic field in the cooling region was less than 2 nT. A cryogenic magnetometer was used to measure the remanent magnetization.

The results were analyzed by using Fisher statistics. In addition, each sample was inspected in orthogonal demagnetization diagrams (As, 1960), where linear trends indicate the presence of a discrete component of magnetization. The direction of magnetization for each sample is defined as the direction of the sum of vectors over a selected temperature interval. For most sites, the two methods gave pole positions within a few degrees of each other. For the two sites in the La Joya Formation and one in the La Boca Formation, a pole position could be obtained only by the vector trend method. The pole positions are listed in Table 1 and depicted in Figure 2, together with the North American pole positions (Irving, 1979). Included are two pole positions from northeastern Mexico by Nairn (1976) and one from southern Mexico by Guerrero (1975), all of Mesozoic age.

## DISCUSSION

All pole positions are shown in the Pacific hemisphere because this yields the simplest polar-wander path. It should be recognized that the polarity of the remanent magnetization for these rocks is unknown. We did collect 41 samples from the Permian Guacamaya Formation in the Peregrina Canyon, in order to establish a known reversed polarity, but we could not extract any useful information from the samples.

The large structural corrections for sites 1 through 9 may conceal a magnetic overprint in the data imparted by the present field or during the Laramide deformation. To test for the possibility of magnetic overprints, a comparison was made between both the natural remanent magnetization (NRM) and demagnetized directions with and without structural corrections (Fig. 3). A recent remagnetization can clearly be excluded. For the remaining sites, 10 through 17, the structural corrections are small. Only the youngest three sites (15, 16, 17) could possibly have been recently remagnetized (Fig. 3, bottom).

The curve in Figure 3 depicts the expected direction through time for the sampling sites had they been rigidly connected to North America (calculated from the data compilation of Irving, 1979). With the exception of the youngest sites (15, 16, 17), none of the observed directions coincides with this curve, and we can safely conclude that no overprint—present, Laramide, or any other time—is concealed in the data. A prefolding age of magnetization is also demonstrated by a positive fold test for sites 1 through 5. The bedding attitudes vary from 35° dip, 159° dip direction for site 1 to 80°, 051° for sites 2, 3, and 4, and 79°, 044° for site 5. Only after structural corrections do the pole positions agree (Fig. 3, top). Finally, the occurrence of reversals (sites 1 through 4 versus site 5, sites 6, 7, and 8 versus site 9, site 15 versus site 16) and mixed polarities (sites 16, 17) provides further evidence that the primary magnetization has been isolated for all sites.

The pole positions from the Huizachal Group (pole 1-14, Fig. 2) fall into three clusters. We interpret this to indicate that the Huizachal Group may comprise three distinct units rather than the two so far recognized, the La Boca and La Joya For-

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mations. The existence of three units is consistent with field evidence, although it has not yet been demonstrated (Belcher, 1979).

The more significant finding is that pole positions from Mexico differ dramatically from the Mesozoic results of North America. Before discussing the tectonic implications of these data it is necessary to establish whether the results apply to all or at least a large part of Mexico or whether they reflect local rotations. A rigorous answer can be obtained only by additional sampling over a much larger area, but this is hampered by the extensive Cretaceous and Tertiary cover. Some basic structural considerations strongly suggest that the paleomagnetic results apply to at least northeastern Mexico.

The sampling sites are located on three early Mesozoic horst blocks identified by Belcher (1979). The horst blocks are now a structural part of the Sierra Madre Oriental, a mountain range formed during the early Tertiary Laramide orogeny (Fig. 1). It appears that the horsts and other positive structures influenced the Laramide folding. Folds developed in the platform sediments deposited over horst blocks are broad, low-amplitude features, whereas the folds of basinal sediments deposited in adjacent areas are tightly appressed, high-amplitude structures. The consistency of the paleomagnetic data suggests that the horst blocks did not undergo differential rotations. Site 1 on the Miquihuana block yields a pole position similar to that of sites 2, 3, and 4, which are located on the Ciudad Victoria–Huizachal block. The same is true for site 12 (Galeana block) as compared to sites 10, 11, 13, and 14 (Ciudad Victoria–Huizachal block). The internal consistency of data from these horst blocks indicates that a large part of northeastern Mexico acted as one structural unit.

Paleomagnetic results from other areas

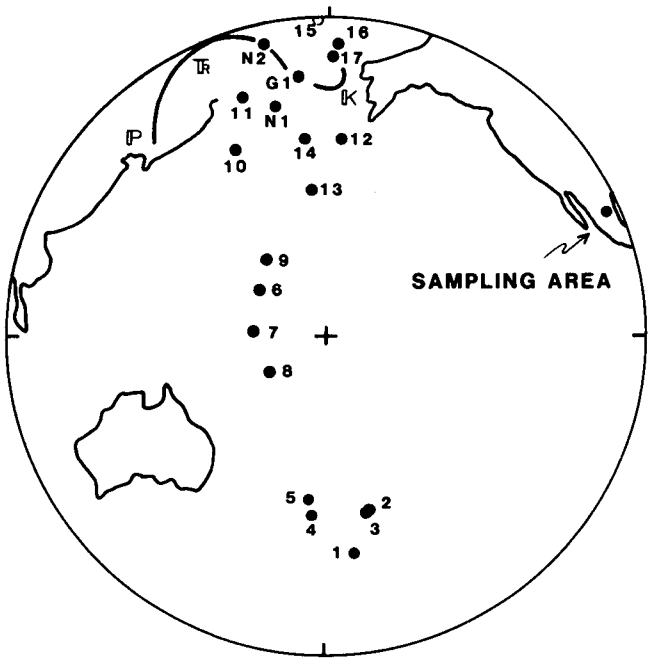



**Figure 2. Mesozoic pole positions from Mexico. Numbering as in Table 1. Solid curve delineates polar-wander path for stable North America (after Irving, 1979) from Permian through Cretaceous time.**

TABLE 1. STATISTICAL PARAMETERS OF PALEOMAGNETIC ANALYSIS OF SAMPLES FROM NORTHEASTERN MEXICO

| | Site | Treat | N | R | D | I | K | α95 | Lat | Long | dp | dm |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| La Boca Fm | | | | | | | | | | | | |
| (1) Miquihuana | 23.7/260.2 | TD650 | 28/5 | 26.5 | 30.0 | 17.3 | 18.4 | 6.5 | 57.7S | 192.5 | 3.5 | 6.8 |
| (2) Caballeros Canyon | 23.8/260.9 | TD650 | 14/6 | 12.7 | 40.8 | 3.1 | 10.2 | 13.1 | 44.7S | 194.1 | 6.6 | 13.1 |
| (3) | | TD650 | 15/2 | 14.5 | 40.0 | 3.2 | 26.6 | 7.5 | 45.4S | 194.1 | 3.8 | 7.6 |
| (4) | | TD650 | 20/4 | 18.3 | 44.7 | 27.7 | 11.4 | 10.1 | 47.0S | 174.7 | 6.0 | 11.1 |
| (5) Peregrina Canyon | 23.8/260.8 | vector | 12 | 11.0 | 229.4 | -25.1 | 10.7 | 13.9 | 42.2S | 174.4 | 8.1 | 15.0 |
| (6) La Boca Canyon | 23.8/260.8 | TD650 | 18/5 | 17.5 | 284.1 | - 6.4 | 34.4 | 6.0 | 11.5N | 162.1 | 3.0 | 6.0 |
| (7) | | TD650 | 17/3 | 16.6 | 275.9 | -18.0 | 41.5 | 5.6 | 1.6N | 160.1 | 3.0 | 5.8 |
| (8) | | TD650 | 14/1 | 13.5 | 263.8 | -18.2 | 25.6 | 8.0 | 9.4S | 164.8 | 4.3 | 8.3 |
| (9) | | TD650 | 19/6 | 17.7 | 110.8 | - 1.2 | 14.2 | 9.2 | 19.2N | 162.8 | 4.6 | 9.2 |
| (10) Huizachal Canyon | 23.6/260.8 | TD625 | 49/2 | 47.7 | 141.1 | - 4.6 | 35.5 | 3.5 | 46.8N | 147.1 | 1.7 | 3.5 |
| (11) | | TD625 | 39/0 | 36.2 | 156.1 | -13.3 | 13.5 | 6.5 | 61.5N | 138.4 | 3.4 | 6.6 |
| (12) Galeana | 24.7/259.9 | TD600 | 17/0 | 16.6 | 137.1 | -47.9 | 36.1 | 6.0 | 51.7N | 186.1 | 5.1 | 7.9 |
| La Joya Fm | | | | | | | | | | | | |
| (13) Peregrina Canyon | 23.8/260.8 | vector | 14 | 12.8 | 124.3 | -34.0 | 10.9 | 12.6 | 38.1N | 176.4 | 8.2 | 14.4 |
| (14) Huizachal Canyon | 23.6/260.8 | vector | 13 | 12.4 | 141.5 | -23.3 | 19.1 | 9.7 | 51.8N | 160.3 | 5.5 | 10.4 |
| Zuloaga Fm | | | | | | | | | | | | |
| (15) Peregrina Canyon | 23.8/260.8 | TD300 | 16/5 | 15.9 | 180.8 | -37.5 | 263.5 | 2.3 | 87.1N | 66.2 | 1.6 | 2.7 |
| (16) Huizachal Canyon | 23.8/260.8 | TD400 | 10/3 | 9.8 | 349.9 | 44.4 | 42.8 | 7.5 | 80.5N | 188.0 | 5.9 | 9.4 |
| Cuesta del Cura Fm | | | | | | | | | | | | |
| (17) Peregrina Canyon | 23.8/260.8 | TD400 | 19/6 | 18.9 | 345.1 | 43.5 | 144.2 | 2.8 | 76.4N | 180.5 | 2.2 | 3.5 |
| Huizachal Group | | | | | | | | | | | | |
| N1 La Murella | 26.3/258.6 | TD500 | 29 | 27.7 | 330.4 | 28.9 | 22 | 5.9 | 60.6N | 154.8 | 3.6 | 6.4 |
| Nazas Fm | | | | | | | | | | | | |
| N2 Los Cerritos Colo. | 25.5/254.5 | TD500 | 14 | 13.5 | 350.6 | 28.4 | 24 | 8.2 | 76.4N | 119.2 | 4.9 | 9.0 |
| San Ricardo Fm | | | | | | | | | | | | |
| G1 Cintalapa | 16.8/266.3 | TD630 | 89/12 | 85.5 | 340.3 | 19.9 | 25.4 | 3.0 | 69.8N | 160.0 | 1.7 | 3.2 |

Note: Site = north latitude/east longitude of sampling site; Treat = thermal demagnetization at specified temperature; N = number of samples included in the statistics/number of samples rejected; R = resultant vector; D = declination; I = inclination; K = precision parameter; α95 = 95% circle of confidence; Lat, Long = paleomagnetic poleposition; dp, dm = semi-angles of ellipse of confidence. Entry N1, N2 from Nairn (1976), entry G1 from Guerrero (1975).

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in Mexico are rare, and more important, there are no results for early Mesozoic rocks. Nairn (1976) sampled undifferentiated Huizachal red beds, near La Murella, some 200 km north of our northernmost sampling site, but these are assigned a Jurassic age (Imlay and others, 1948). Nairn sampled another site near Torreon, 340 km west of our sampling area, in the Nazas Formation. The Nazas is interlayered with volcanic rocks, dated at 126 ± 16 m.y. (Halpern and others, 1974), latest Jurassic to Early Cretaceous. The only data from southern Mexico were obtained from the Upper Jurassic–Lower Cretaceous San Ricardo Formation in Chiapas (Guerrero, 1975). These three pole positions (see Fig. 2) agree well with our northeastern Mexican poles of similar age. This implies that most of Mexico, including the Yucatan block, was part of a single coherent plate by Early Cretaceous time.

Because the relative ages of the various Huizachal outcrops are poorly constrained, only a general comparison between the Mexican and cratonic North American pole positions is possible. At the beginning of the Mesozoic, the sampling area was in a position rotated some 130° clockwise relative to North America and then gradually rotated counterclockwise until about the Middle Jurassic. Throughout this time, there is no evidence of a major latitudinal change relative to North America. The pole positions from the Zuloaga Limestone and the Cuesta del Cura Formation imply, at face value, that some slight differential motion continued into the Cretaceous, but the current Cretaceous data base is insufficient for any more than speculation. It is interesting to note, though, that Urrutia and Pal (1977) interpreted their paleomagnetic data on Oligocene volcanic rocks from the Mexican volcanic belt to imply counterclockwise rotation relative to North America.

In recent paleogeographic reconstructions (for example, Van der Voo and others, 1976; Pilger, 1978; Walper, 1980), South America closely abutted North America at the onset of the Mesozoic, leaving no space for Middle America. The data presented here offer two possible solutions. If our results are representative for most of Mexico, then Mexico was not a structural part of North America. Although paleomagnetic data do not yield information about paleolongitude, it is probable that the large rotational motion of Mexico also had an east-west component, thus allowing the closure of the Gulf of Mexico without creating topological problems.

An alternative interpretation appeals to the possible existence of a set of major faults. De Cserna (1976) proposed three west-northwest–trending transcurrent faults in northern Mexico with left-lateral displacement (near Torreon, Zacatecas, and Mexico City). Our sampling area falls between the two northern faults. The sense of motion is such that it might well cause a counterclockwise rotation of small crustal blocks. Our data could readily be explained by this model. This scenario is very similar to the tectonic history of the Pacific Northwest (see Beck, 1980, for a review). De Cserna (1976) postulated a late Paleozoic–early Mesozoic motion along these faults; this is the same time span in which the paleomagnetic data imply the rotation.

At this point, there is no compelling reason to prefer one interpretation over the other. It is important, however, to realize that in either case the data require a major fault in northern Mexico. A megashear in this area has been proposed by several other workers (for example, Walper and Rowett, 1972; Walper, 1980; Silver and Anderson, 1974; Anderson and Schmidt, 1980; Pilger, 1978; Van der Voo and others, 1976), but our data, we believe, are the strongest evidence so far.

A second inference that can be drawn from our results and the results from southern Mexico (Guerrero, 1975) is that all of Mexico seems to have reached its present geometric relationship with respect to North America by Middle to Late Jurassic time, thereby establishing the Gulf of Mexico. This interpretation is in full agreement with stratigraphic studies (Salvador and Green, 1980) and seismic studies (Buffler and others, 1980, 1981; Ibrahim and others, 1981) that demonstrate that the Gulf of Mexico formed in early Mesozoic time and obtained its present outline in the Late Jurassic.

While there is some broad consensus on the early Mesozoic history of Mexico, the details are still poorly understood. Did Mexico consist of one or several tectonic blocks? Where was Mexico located at the beginning of the Mesozoic? Where exactly is the discontinuity between Mexico and North America? Clearly, these questions require extensive additional work.

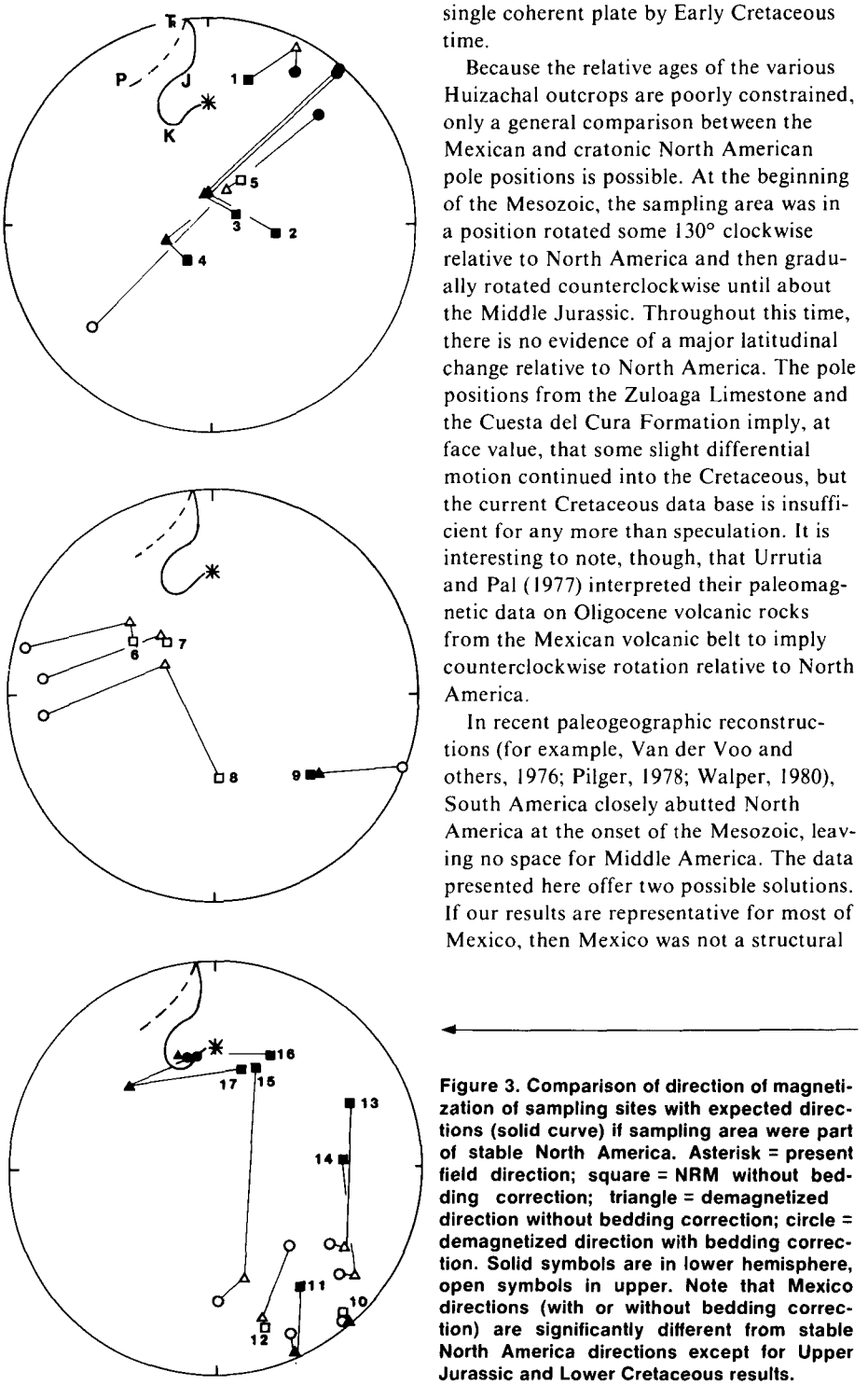



**Figure 3. Comparison of direction of magnetization of sampling sites with expected directions (solid curve) if sampling area were part of stable North America. Asterisk = present field direction; square = NRM without bedding correction; triangle = demagnetized direction without bedding correction; circle = demagnetized direction with bedding correction. Solid symbols are in lower hemisphere, open symbols in upper. Note that Mexico directions (with or without bedding correction) are significantly different from stable North America directions except for Upper Jurassic and Lower Cretaceous results.**

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## ACKNOWLEDGMENTS


Reviewed by R. T. Merrill and R. H. Pilger. Supported by the Shell Development Company, Sloan Foundation, and National Science Foundation Grant EAR-77-23347. We thank R. T. Buffler, F. J. Shaub, D. K. Swartz, and M. M. Testarmata for stimulating discussions and in-house reviews. As of June 1981, the Galveston Geophysics Laboratory has been reorganized into the Institute for Geophysics. Contribution No. 505.

MANUSCRIPT RECEIVED JUNE 15, 1981
REVISED MANUSCRIPT RECEIVED AUG. 31, 1981
MANUSCRIPT ACCEPTED NOV. 12, 1981